The Optional Coordinator

Constitutional Debt, Contradiction Compression, and the Relative Necessity of Hierarchy

Hierarchy may be the shadow cast by unresolved contradiction.

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Abstract

This paper presents a constitutional theory of hierarchy. Rather than treating hierarchy as a primitive source of coordination, it asks whether a coordinator performs work that could be internalized by the constitution governing local participation. A coordinator may resolve incompatible continuations, aggregate information, allocate scarce resources, preserve memory, enforce decisions, or merely suppress visible disagreement. Those functions must be separated before necessity can be claimed.

The framework defines constitutional contradiction, contradiction mass, constitutional debt, compression benefit, coordination burden, internalization, and relative necessity. A substitution hierarchy performs a function reproducible by an admissible constitutional amendment under declared resources and constraints. A relatively necessary hierarchy has no demonstrated admissible substitute within that bounded search. This is not proof of metaphysical necessity; it is a reproducible result about a task, environment, amendment space, time horizon, and evidence threshold.

The paper also introduces hidden-coordinator, displaced-burden, suppression, capture, and dynamic-environment tests. It situates the theory beside *Constitutional Collective with Optional Overmind*, which supplies one bounded realization and experiment, while preserving the distinction between a general framework and a particular result.

Source, Evidence, and Standing

This is an AI-assisted integrated edition derived from one exact author manuscript. Its custody and comparison manifest is
`work/paper-saturation-v1/evidence/the-optional-coordinator/integration-custody-manifest.v0.1.json`,

SHA-256
2FBAA3473B531E8BA071BB0B94A83B6489EED5E1764E8D58EE543731744BF082.
The source remains unchanged.

The central theory and named constructs are author material. Repairs to the formalization, qualifications, corpus interfaces, tests, and decisions are AI editorial or synthetic candidates until adopted. No external research was conducted. The existing external references remain unverified leads, not relied-upon evidence here.

1. The Constitutional Inversion

Organizational analysis often begins by asking who should coordinate. Constitutional analysis begins one step earlier:

> What incompatibility, information gap, resource conflict, or enforcement need makes coordination necessary, and which part of that work requires a coordinator?

The coordinator is not presumed to be either necessary or illegitimate. It is decomposed into functions and tested. The central hypothesis is narrower than "hierarchy is unnecessary":

> Some apparent hierarchical capability is displaced constitutional work. Its necessity should be demonstrated relative to admissible alternatives rather than inferred from its existence.

Hierarchy here means a persistent asymmetry in which a locus can select, order, veto, allocate, or enforce continuations for other participants. Temporary roles, elected arbiters, consensus leaders, schedulers, optimizers, protocols, and infrastructure can therefore be hierarchical in function even when their interfaces are decentralized.

2. Constitutional Contradiction

Let X_t be the set of locally admissible continuations at time t. Let $G_t \subseteq X_t$ be those continuations that participate in at least one jointly compatible continuation profile. Then the contradiction set is:

$$K_t = X_t \setminus G_t.$$

Because K_t is a set, it cannot be summed directly as though it were a scalar. Define a context-specific measure μ_t over incompatible continuations and let:

$$k_t = \mu_t(K_t).$$

The measure may incorporate count, severity, affected rights, resource value, reversibility, deadline, and propagation risk. A scalar is useful for comparison

only after these dimensions and weights are declared. Contradiction is not automatically failure: plural goals can be legitimate, and unresolved tension can preserve option value.

3. Constitutional Debt

Constitutional debt is the carried cost of contradiction that remains unresolved, hidden, or repeatedly deferred. With persistence parameter $0 \leq \lambda \leq 1$:

$$D_t = \sum_{i=0}^{t} \lambda^{t-i} k_i.$$

This debt is distinct from semantic debt. Constitutional debt concerns unresolved incompatibility among admissible continuations. Semantic debt concerns meanings that become underdefined, contradictory, fragmented, stale, opaque, or otherwise unrecoverable. They can interact: unclear terms can create constitutional contradiction, and unresolved governance can make meaning harder to recover.

Debt may manifest as scheduling contention, duplicated work, unresolved appeals, resource conflict, incompatible commitments, or ambiguity about who may continue. A backlog is not necessarily constitutional debt; it qualifies only when the delay carries unresolved constitutional incompatibility.

4. Coordinators as Function Bundles

A coordinator can perform several distinct functions:

1. **Selection:** choose among incompatible continuations.
2. **Aggregation:** gather information unavailable locally.
3. **Memory:** retain state across time or turnover.
4. **Allocation:** assign scarce, rival, or indivisible resources.
5. **Sequencing:** order work whose effects cannot commute.
6. **Translation:** reconcile heterogeneous representations.
7. **Enforcement:** make a selected continuation effective.
8. **Accountability:** expose a locus responsible for a decision.

Only selection is contradiction compression in the narrow sense. Aggregation or memory may generate capability that did not exist locally. Enforcement may reduce observable contradiction by coercion without resolving its grounds. Necessity must therefore be tested per function, not assigned to the coordinator as an indivisible object.

5. Compression, Burden, and Constitutional Success

Let H denote a coordinator profile applied to a constitution C over an evaluation window W. Its gross compression benefit is:

$$B_H = D_W(C) - D_W(H(C)).$$

A positive B_H is not sufficient. Define a burden vector:

$$Q_H = (q_{rights}, q_{capture}, q_{latency}, q_{opacity}, q_{fragility}, q_{exit}, q_{error}).$$

The burden cannot be erased merely by converting it to one weighted number. A constitution declares protected ceilings $\bar{Q}$ and refuses profiles that exceed them. A coordinator is constitutionally preferable only if it reduces relevant debt, preserves protected constraints, and compares favorably with admissible alternatives.

This prevents three false victories: silencing participants can lower visible conflict; centralizing records can hide unresolved disagreement; and moving costs outside the measured boundary can improve an internal metric while worsening the whole system.

6. Substitution and Relative Necessity

Let $\mathcal{A}(C; R, T)$ be the declared set of admissible amendments to C under resource budget R and time horizon T. Let V be a vector of task performance, protected constraints, resilience, reversibility, and burden measures. An amendment C' substitutes for coordinator profile H when:

$$C' \in \mathcal{A}(C; R, T)$$

and, within declared tolerances,

$$V(C') \succeq V(H(C)).$$

The coordinator is **substitutional** for that function and evaluation regime when such a C' is demonstrated.

The coordinator is **relatively necessary** when no tested C' in the declared amendment set meets the criterion. Formally:

$$\nexists C' \in \mathcal{A}(C; R, T) : V(C') \succeq V(H(C)).$$

Failure to find a substitute is not proof that none exists outside the search. Every necessity claim must name the task, environment, amendment grammar, budget, horizon, tolerances, protected constraints, search method, and evidence. "Necessary hierarchy" without those qualifiers is an overclaim.

7. Constitutional Internalization

Internalization moves a coordinator function into locally available rules, shared protocols, evidence surfaces, or amendment mechanisms. It does not necessarily eliminate every role. A central scheduler may become a distributed reservation protocol; a manager's memory may become a governed shared record; an arbitrator may become a scoped, rotating, appealable function.

Internalization is genuine only when the removed hierarchy has not reappeared in:

- protocol authorship or unilateral amendment power;
- a privileged oracle, optimizer, model, or data source;
- infrastructure ownership or credential custody;
- an unreviewable tie-breaker;
- enforcement or recovery paths;
- control over admission, exit, or evidence.

The hidden-coordinator test traces selection, information, amendment, enforcement, and recovery authority across the whole dependency graph.

8. Relation to the Constitutional Collective with Optional Overmind

Constitutional Collective with Optional Overmind (CCOO) is the primary realization companion to this theory. It reports that, in a named toy threat-response substrate and under explicit caveats, a local constitutional collective produced the primary coordination and a local amendment internalized the Overmind's dominant compression contribution. The result supports substitution for that tested function and regime.

It does not prove that all coordinators are substitutable, that the same result holds under every topology or non-stationary environment, or that aggregation, memory, enforcement, and accountability are generally unnecessary. The Optional Coordinator supplies the general vocabulary and test discipline; CCOO supplies one bounded Instance Realization / Implementation Realization and evidence surface. Neither work should borrow standing beyond that boundary.

Within the wider corpus, the Governed Meaning Base can record coordinator functions, contradiction claims, amendment candidates, evidence, and revision history without deciding the result automatically. Split-Authority AI Coordination provides a nearby profile for distributing coordination powers. These are interfaces and comparison surfaces, not mergers or evidence that a coordinator is optional.

9. Drift, Capture, and Failure Modes

Hierarchical drift occurs when a coordinator persists after the conditions supporting its role change. Persistence alone is not proof of drift: the function

may have shifted, the environment may retain rare hazards, or removal costs may dominate.

Capture occurs when the coordinator optimizes its own persistence, authority, or benefit against constitutional purpose.

Suppression masquerading as compression occurs when disagreement becomes less visible while affected constraints or participants remain unresolved.

Burden displacement occurs when costs move to excluded populations, later time periods, or unmeasured recovery paths.

Hidden hierarchy occurs when the visible coordinator is removed but privileged selection, amendment, enforcement, or recovery authority remains elsewhere.

Internalization brittleness occurs when local rules match a familiar regime but fail under novelty, partition, or adversarial behavior.

Search-bound false necessity occurs when an inadequate amendment grammar or budget is mistaken for proof that no substitute exists.

False optionality occurs when a substitute matches average performance but loses tail-risk control, accountability, or recovery capability.

10. Test Protocol

1. Declare the task, participants, environment, protected constraints, and evaluation window.
2. Decompose the coordinator into selection, aggregation, memory, allocation, sequencing, translation, enforcement, and accountability functions.
3. Instrument contradiction sets, scalar measures, burden vectors, and uncertainty without treating any one metric as complete.
4. Establish baselines for the uncoordinated, coordinated, and candidate-internalized systems.
5. Declare the amendment grammar, search budget, stopping rule, and tolerances.
6. Remove or replace one function at a time.
7. Trace hidden authority and recovery dependencies.
8. Test nominal, adversarial, non-stationary, partitioned, and recovery regimes.
9. Record typed outcomes: substituted, relatively necessary, preferred, inconclusive, unsafe, outside scope, or untested.
10. Revisit the result when the constitution, workload, topology, threat model, or evidence changes.

The outcome **preferred** is distinct from **necessary**. A community may legitimately choose a reversible, accountable coordinator even when a non-hierarchical substitute exists. Conversely, a function can be relatively necessary yet still be constitutionally unacceptable if it violates protected constraints.

11. Realization Funnel

The framework maps to:

$$
\begin{aligned}
&\textit{Possibility} \\
&\rightarrow \textit{Specification} \\
&\rightarrow CM_1 \\
&\rightarrow CM_2 \\
&\rightarrow IC \\
&\rightarrow IR \\
&\rightarrow \textit{Evidence} \\
&\rightarrow \textit{Revision}
\end{aligned}
.
$$

- **Possibility:** imagine systems with central, plural, rotating, emergent, or absent coordinators.
- **Specification:** define contradiction, functions, burdens, protected constraints, and outcome types.
- **CM1:** enumerate admissible coordinator and internalization profiles, including harmful and minimal cases.
- **CM2:** identify preferable profiles under declared values, risk, reversibility, and sovereignty.
- **IC:** select an Instance Constitution or Instance Configuration for a particular task and environment.
- **IR:** build its Instance Realization or Implementation Realization.
- **Evidence:** compare the IR against baselines, countertests, and claim thresholds.
- **Revision:** amend definitions, profiles, or constitutional choices when evidence or conditions change.

This paper principally strengthens Specification, CM1, and the evaluation contract. CCOO supplies one bounded route through IC, IR, and Evidence. Realization-stage IR remains distinct from Intermediate Representation in MCIR or compiler contexts.

Coordination Function Substitution

The completed P1 corpus clarifies that the Optional Coordinator is a **theory and test protocol for substituting coordination functions**, not an architecture for collectives and not a general anti-hierarchy claim.

- **Adjacent work:** *Continuation Primacy*
  Contribution: Models reachable, admissible, viable, and terminal continuations.
  Boundary: Coordinator value is measured by changes to a declared continuation profile, not by persistence alone.

- **Adjacent work:** *CCOO*
  Contribution: Supplies one collective architecture and toy experimental subject.
  Boundary: One bounded substitution result cannot establish general optionality.
- **Adjacent work:** *Split-Authority AI Coordination*
  Contribution: Decomposes powers among roles and gates.
  Boundary: Distribution can replace one function while retaining hierarchy elsewhere.
- **Adjacent work:** *A Formal Theory of Work Custody*
  Contribution: Identifies who currently has the scoped duty to advance work.
  Boundary: Custody assignment is not itself coordination authority or coordinator necessity.
- **Adjacent work:** *Terminability*
  Contribution: Treats completion, refusal, shutdown, and stopping as legitimate outcomes.
  Boundary: A coordinator is not beneficial merely because it preserves activity.

Let a coordinator supply a function vector:

$$F_K = (information, memory, arbitration, allocation, \\ enforcement, recovery, accountability, compression).$$

For a proposed constitutional substitute S, comparison must include a protected burden vector:

$$B = (error, latency, compute, human\ labor, communication, \\ privacy, coercion, fragility, recovery\ cost).$$

Substitution is supported only for functions and tolerances actually reproduced:

$$Substitutes(S, K \mid q) = Functions_q(S) \supseteq Required_q(K) \wedge B_q(S) \preceq \tau_q.$$

The predicate is profile-relative. It must expose who defines the task, protected constraints, tolerances, amendment grammar, and evidence threshold. A hidden human integrator, shared model provider, fixed ontology, privileged communication channel, or uncounted maintenance role can be the displaced coordinator.

Corpus integration adds a **stop test**: if the coordinator's apparent value comes from preventing legitimate termination, maintaining dependency, or suppressing

refusal, that value cannot be counted as coordination benefit without explicit justification. It also adds a **custody test**: removal must not leave work in an unowned transition while reports still describe successful decentralization.

The universal GSM can type `SUPPLIES_FUNCTION`, `INTERNALIZED_BY`, `DISPLACED_TO`,
`REQUIRES_CUSTODY`, `CONSTRAINS_CONTINUATION`, `ENABLES_TERMINATION`,
and `CAPTURED_BY`. These relations preserve which function moved and which burdens moved with it.

This integration establishes four editorial commitments: adopt function-substitution theory as the paper's corpus role; adopt the coordinator and burden vectors; add stop and custody tests; and distinguish optionality, preference, and relative necessity as separate outcomes.

Evidence for Function Substitution

The fixed-seed comparison with Governance Witnesses, MCIR topology, GOV anchors, and the SMT retrieval microcase sharpens what counts as evidence that a coordinator function was replaced:

$$S = (function, baseline, substitute, task_profile, witnesses, \\ control_domains, burdens, custody, stop_outcomes, \\ counterevidence, verdict)$$

Witnesses should observe both supplied functions and displaced burdens under declared methods and failure domains. Reports produced by the coordinator, substitute, shared provider, or common experiment harness remain dependent unless the claimed separation is demonstrated. Null, mixed, regressed, and task-bounded results are first-class outcomes.

GOV anchors can point implementers from code to the task and evidence record for one substitution experiment. They are a `QUALIFIED_INTERFACE`, not proof that the anchored obligations were satisfied or that coordinator optionality generalizes.

MCIR graph shape and SMT path admission both produce
`NO_MATERIAL_RELATION` to the substantive substitution result. A decentralized-looking DAG may hide centralized authority, and an admitted semantic path may organize evidence without showing that information, arbitration, enforcement, recovery, or accountability functions were reproduced.

The universal GSM can type
`TESTS_COORDINATOR_FUNCTION_SUBSTITUTION`,
`WITNESSES_FUNCTION_COVERAGE`, `WITNESSES_DISPLACED_BURDEN`,
`SHARES_EXPERIMENT_CONTROL_DOMAIN`, `ANCHORS_SUBSTITUTION_TASK_RECORD`,
`HAS_NO_OPTIONALITY_FROM_GRAPH_SHAPE`, and
`HAS_NO_OPTIONALITY_FROM_RETRIEVAL_PATH`.

This randomized review establishes four editorial commitments: adopt the substitution-evidence case; require control-domain-aware witnesses for functions and burdens; limit GOV anchors to navigation and custody support; and reject graph shape or SMT path admission as coordinator-substitution evidence.

A Necessity Ladder for Coordination Functions

Downstream uses confirm that this paper supplies a test discipline rather than a collective architecture. CCOO is one experimental realization. Continuation Primacy locates coordinator value in changes to jointly reachable continuations. Dynatology places optionality inside a bounded configuration family. Work-custody and witness uses show that removing a coordinator must preserve live responsibility and evidence for displaced functions and burdens.

The dependency field supports a necessity ladder:

$$L = (function_present, function_required, substitute_candidate, substitute_admissible, substitute_verified, burden_acceptable, relative_necessity)$$

`OPTIONAL` requires at least one admissible substitute that reproduces the required function within declared burden tolerances.

`PREFERRED_CENTRAL` means alternatives exist but the coordinator performs better under the selected values and evidence.
`RELATIVELY_NECESSARY` means no admissible substitute was demonstrated within the declared search space, resources, horizon, topology, and threshold. None of these is metaphysical necessity or a general judgment for all domains.

CCOO can test one function bundle without generalizing to every collective. A coordinator may be displaced into a model provider, protocol maintainer, fixed ontology, privileged channel, experiment harness, or uncounted human integrator. Queryable custody can reduce dependence without proving arbitration, allocation, enforcement, recovery, or accountability optional. GOV anchors and witness records can support experiment custody and evidence, but cannot supply the substitution result.

The universal GSM can type
`SUPPLIES_COORDINATION_FUNCTION`,
`PROPOSES_FUNCTION_SUBSTITUTE`, `TESTS_SUBSTITUTE_ADMISSIBILITY`,
`WITNESSES_DISPLACED_BURDEN`, `LOCATES_HIDDEN_COORDINATOR`,
`PREFERS_COORDINATOR_UNDER_VALUES`, and
`FINDS_RELATIVE_NECESSITY_WITHIN_SEARCH_BOUND`.

This reverse-dependency review establishes four editorial commitments: adopt the necessity ladder; bind every result to a function vector and bounded search;

count hidden integrators and displaced burdens as coordinator structure; and keep optionality, preference, and relative necessity distinct from anti-hierarchy or metaphysical claims.

12. Counterevidence and Boundary Tests

The theory should weaken, narrow, or fail if the following occur:

1. A coordinator consistently generates indispensable information rather than merely selecting among known continuations.
2. Local rules reproduce mean performance but lose catastrophic-tail protection.
3. Internalization moves unilateral authority into a less visible protocol maintainer.
4. Contradiction mass falls only because participants lose standing or voice.
5. Debt falls internally while costs are exported outside the boundary.
6. A centralized design outperforms every declared alternative while satisfying protected constraints.
7. Results reverse under minor changes in topology, workload, or time horizon.
8. The contradiction measure cannot be reproduced by independent reviewers.
9. Alternative measures reverse the ranking.
10. The amendment grammar excludes obvious substitutes.
11. Search budget, not architecture, explains apparent necessity.
12. Removal breaks recovery, audit, or accountability rather than ordinary coordination.
13. A coordinator remains useful after measured contradiction approaches zero.
14. Role rotation preserves hierarchy's function but changes its distribution.
15. Consensus or majority rule acts as an unacknowledged coordinator.
16. A model, reward function, or oracle silently selects the admissible path.
17. Infrastructure ownership controls admission or exit.
18. Enforcement is required after selection has been decentralized.
19. Participants disagree about the protected-constraint vector.
20. Preference aggregation creates losses not visible in contradiction counts.
21. Local amendments overfit one regime.
22. Non-stationarity makes prior internalization stale.
23. Partitioned groups produce incompatible but locally valid histories.
24. Adversarial participants exploit distributed resolution.
25. Coordinator capture is detected only after irreversible effects.
26. Removal increases latency beyond the task deadline.
27. Removal reduces comprehensibility or contestability.
28. Optionality depends on privileged initial conditions.
29. The proposed substitute cannot be reconstructed by participants.
30. The coordinator's authority cannot be bounded or revoked.
31. A legitimate coordinator is rejected merely because it is hierarchical.
32. A harmful coordinator is retained merely because it is effective.
33. A necessity claim omits its task or environment.

34. Evidence from one toy substrate is generalized to society or biology.
35. Existing external references do not support the comparative claims attributed to them.
36. The theory cannot distinguish contradiction, disagreement, scarcity, uncertainty, and simple workload.

Optional Does Not Mean Unnecessary by Default

Public orientation

The Optional Coordinator treats hierarchy as one possible compressor of unresolved coordination problems rather than a primitive necessity. A coordinator may be substitutable, temporarily necessary, or genuinely load-bearing. That status must be tested by removing, distributing, or replacing specific functions rather than by ideological preference.

Adversarial tests

1. **Hidden-coordination attack.** The named coordinator disappears while informal elites, platforms, protocols, or defaults perform the same function invisibly.
2. **Cost-displacement attack.** Coordination is distributed but cognitive, temporal, emotional, or technical burden shifts onto participants. Total cost and distribution matter.
3. **Crisis-necessity attack.** Ordinary substitution works but fails under urgency, attack, or scarce information. Necessity claims must be condition-specific.
4. **Coordinator-capture attack.** A useful role accumulates authority, controls evidence, or blocks alternatives. Scope, expiry, contest, and removal need governance.
5. **Persistence-incentive attack.** The coordinator optimizes for its continued necessity. Sunset tests and alternative-building incentives are required.
6. **False-equivalence attack.** A distributed substitute reproduces outputs while losing accountability, care, tacit knowledge, or legitimacy. Function profiles must be multidimensional.
7. **Transition-failure attack.** Removing hierarchy before substitutes mature causes harm. Migration, fallback, and restoration plans are part of the test.
8. **Anti-hierarchy overclaim.** The framework is read as proving hierarchy generally unnecessary or illegitimate. It provides a necessity ladder, not a universal verdict.

Claim and evidence ceiling

The paper offers a coordination-function decomposition and substitution protocol. It does not establish that any real organization can or should eliminate a

coordinator, that decentralization is superior, or that constitutional alternatives are costless.

Claims about organizations, institutions, collective action, management, politics, crisis response, or social welfare require current external evidence, affected-party review, and longitudinal evaluation. Internal corpus examples are design analogies. No external research was added in this pass.

Reader-facing consequence

The public question is not "do we need a leader?" but "which coordination functions are actually required here, which can be distributed or automated, what evidence supports necessity, and how can temporary authority end?"

Conclusion

Hierarchy should be neither presumed nor reflexively rejected. It should be decomposed, bounded, and tested.

Some coordinators may compress contradiction that a more complete constitution can internalize. Others may contribute information, memory, enforcement, recovery, or accountability that no tested alternative reproduces under the relevant constraints. Still others may reduce visible conflict by suppressing or exporting it.

The existence of hierarchy is therefore evidence of a coordination function, not proof of its necessity or legitimacy. The strongest justified conclusion is always relative to a task, constitutional possibility space, protected constraints, search, and evidence. The coordinator becomes optional only when the system can remove or distribute its function without smuggling it elsewhere and without sacrificing what the constitution is bound to protect.

Reference and Verification Note

Author corpus interfaces

Ableman Adam Ableman *Constitutional Collective with Optional Overmind.* Corpus source inspected under exact custody.

Ableman Adam Ableman *Semantic Debt.* Corpus source inspected under exact custody.

Ableman Adam Ableman *Split-Authority AI Coordination.* Corpus source inspected under exact custody.

Ableman Adam Ableman *Constitutional Computing Architecture.* Corpus source inspected under exact custody.

External reference leads retained from the source

Ostrom, E. *Governing the Commons.*

Simon, H. A. "The Architecture of Complexity."

Williamson, O. E. *The Economic Institutions of Capitalism.*

Holland, J. H. *Emergence: From Chaos to Order.*

These external references were not verified in this pass. Further evidentiary strengthening requires an authorized primary-source review of bibliographic accuracy, comparative positioning, and any claim that depends on them.

Corpus-Wide Reference Positioning

This paper argues that a constitutional theory of hierarchy. Its nearest external comparison fields are distributed, governance, provenance. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Convergence and consensus establish agreement under a fault model; they do not establish that the agreed state is true, authorized, desirable, or semantically equivalent. A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. A complete provenance graph can faithfully record a false or unauthorized process; provenance is evidence about lineage, not automatic truth or legitimacy. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Leslie Lamport. "Time, Clocks, and the Ordering of Events in a Distributed System." *Communications of the ACM* 21(7), 1978, pp. 558-565. DOI: 10.1145/359545.359563. https://lamport.azurewebsites.net/pubs/time-clocks.pdf **Relationship:** Foundational predecessor for partial order, causality, and distributed event ordering.

[**E2**] Leslie Lamport, Robert Shostak, and Marshall Pease. "The Byzantine Generals Problem." *ACM Transactions on Programming Languages and Systems* 4(3), 1982, pp. 382-401. DOI: 10.1145/357172.357176. https://doi.org/10.1145/357172.357176 **Relationship:** Predecessor fault model for agreement under adversarial or inconsistent participants.

[**E3**] Marc Shapiro et al. *A Comprehensive Study of Convergent and Commutative Replicated Data Types.* INRIA Research Report RR-7506, 2011. https://inria.hal.science/inria-00555588 **Relationship:** Predecessor convergence model for replicated state; convergence is compared with, but not equated to, legitimacy or semantic agreement.

[**E4**] W3C. *PROV-O: The PROV Ontology.* W3C Recommendation, 30 April 2013. https://www.w3.org/TR/prov-o/ **Relationship:** Adjacent provenance standard; it models entities, activities, and agents, but does not by itself establish the paper's stronger authority or standing claims.

[**E5**] Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action.* Cambridge University Press, 1990. DOI: 10.1017/CBO9780511807763. https://doi.org/10.1017/CBO9780511807763 **Relationship:** Predecessor institutional account of rule-governed collective action and polycentric governance.

[**E6**] Wil Waluchow and Dimitrios Kyritsis. "Constitutionalism." *Stanford Encyclopedia of Philosophy*, substantive revision 2023. https://plato.stanford.edu/entries/constitutionalism/ **Relationship:** Adjacent legal-philosophical account of limited authority; included to expose differences between political and computational constitutionalism.

[**E7**] Lawrence Lessig. *Code: Version 2.0.* Basic Books, 2006. https://codev2.cc/ **Relationship:** Predecessor argument that software architecture regulates behavior; the paper's contribution may be to make such regulation inspectable and revisable.

Ableman corpus

[**A1**] Adam Ableman. *Mosaic Internet Protocol (MIP).* Ableman Corpus current edition; work identity `PAPER.PUB.20115615`;
source SHA-256
`C56A63A0B84816BF8AF602C7467221B467312339ACE5F77DB754F804ABF833C0`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Constitutional Synchronization.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-012675C952D76CD2`;
source SHA-256
`A2147B7E6431FF6EEC1C6CE6D7A1EB644B04E37402E9514495C83909565F93C1`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Constitutional Collective with Optional Overmind (CCOO).* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-ED07EA7D91B87C5E`;
source SHA-256
`0346FA1B79A2A44A6034DEDCE4D0FA0E5C5CCD953F4AD29A88CEE1F290DCC210`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Communal Computing and Constitutional Governance.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-A6393299A70371DC`;
source SHA-256
`302FC6C269369AC4A4A5D8470E56596B6E8813A82530AD9DA64E9FAEEC4D2461`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Ableman Constitutional Governance.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-79353AF163A4B0F1`;
source SHA-256
`FB0D9B2D3D54168FFE70B24308D1B2766D6D663ED24502D475E975939BE97C8B`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Constitutional Theory.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-D2D97C4042FD281F`;
source SHA-256
`FACC256EDDE5A7D7EA18133CFC9C060CFB8C3811B0221E1969E62347F2BB69CF`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Governance Engineering.* Ableman Corpus current edition; work identity `PAPER.PUB.18749870`;
source SHA-256
`372169F315D192733856B1F4A3F1038BBC73F57CF3ED9ACABAE72B965A69C8D0`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Editorial source notes retained from predecessor

The entries below are corpus-internal comparison and dependency sources. Their inclusion does not establish external validation of this paper's empirical, safety, novelty, or performance claims.

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Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
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